

Template for Computer Programming I Practical Recording

Date : 13/02/2025

Problem Specification : Write program to print out that many powers of 2.

Assumption : User inputs are valid.

Limitation : N/A

Input : Input to Number: num

Processing : Calculate num powers of 2.

Output : print value of num powers of 2

Algorithm :
Step1: Prompt "Enter the number"
Step2: Read the number num
Step3: Initialise the power: power ← 1
Step4: Initialise i ← 0
Step5: Repeat Step 6, Step 7, Step 8 until (i < num)
Step6: Print ("2^"+i+"="+power)
Step7: i ← i+1
Step8: power ← power*2
Step9: End

Programme listing : Programme file **attached**

Test data and expected output :
1. Test data: n = 12
Expected output:
2^0=1
2^1=2
2^2=4
2^3=8
2^4=16
2^5=32
2^6=64
2^7=128
2^8=256
2^9=512
2^10=1024
2^11=2048

Output obtained for test data : 1. Test data: n = 12

Obtained output:

- $2^0=1$
- $2^1=2$
- $2^2=4$
- $2^3=8$
- $2^4=16$
- $2^5=32$
- $2^6=64$
- $2^7=128$
- $2^8=256$
- $2^9=512$
- $2^{10}=1024$
- $2^{11}=2048$

Discussion : Take note of how to use for loops.

Conclusion : The user inputs the number. then program processes the input, calculate value of number powers of 2 .then print the number powers of 2.